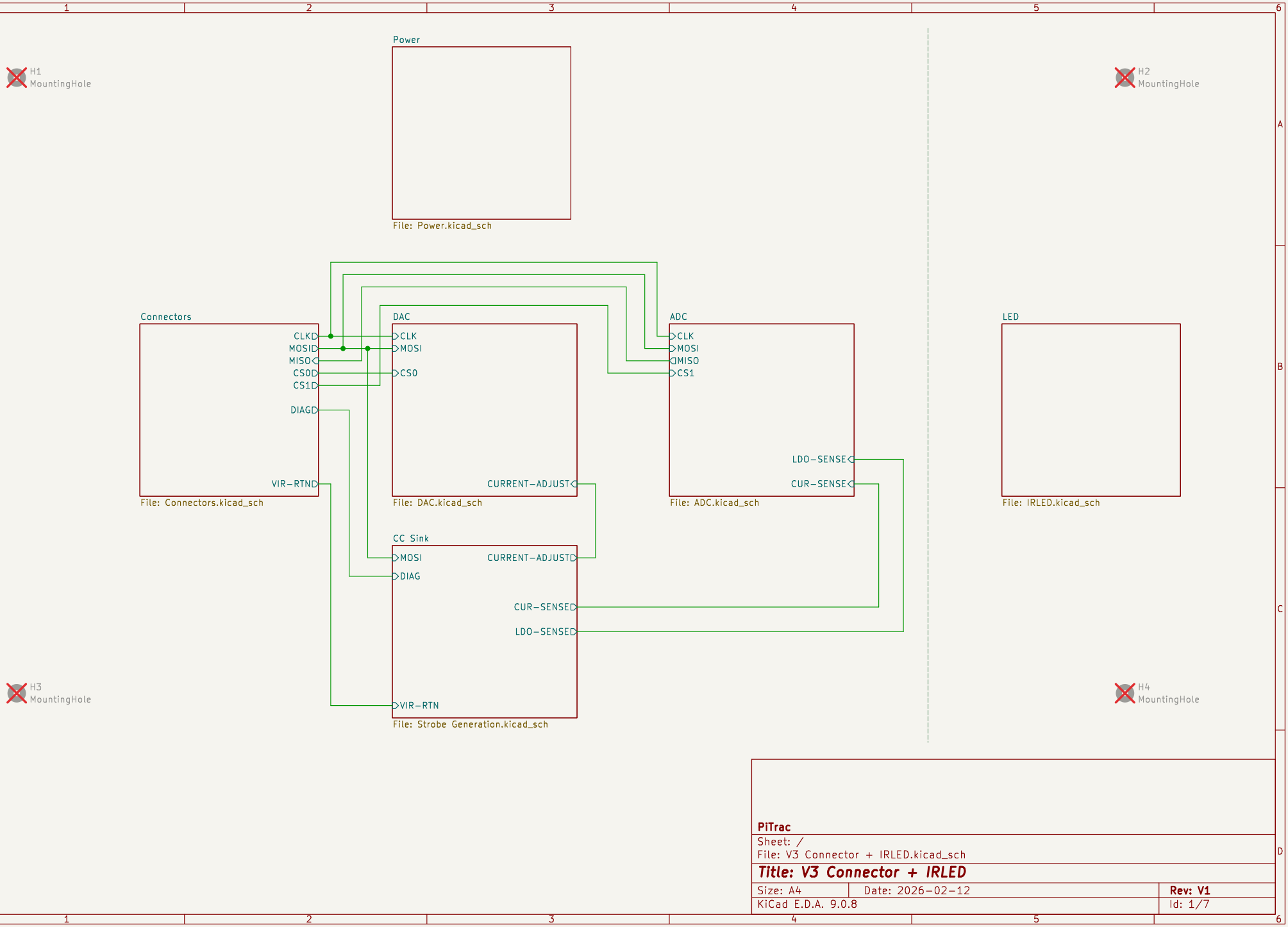
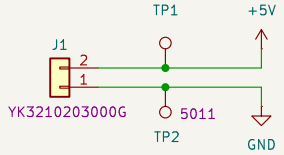
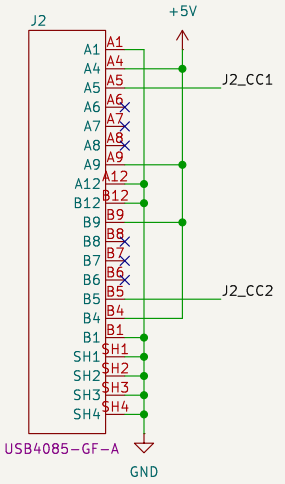
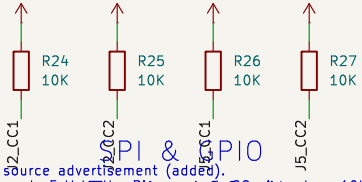

H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
Power
File: Power.kicad_sch
Connectors
CLK
MOSI
MISO
CS0
CS1
DIAG
VIR-RTN
File: Connectors.kicad_sch
DAC
CLK
MOSI
CS0
CURRENT-ADJUST
File: DAC.kicad_sch
ADC
CLK
MOSI
MISO
CS1
LDO-SENSE
CUR-SENSE
File: ADC.kicad_sch
LED
File: IRLED.kicad_sch
CC Sink
MOSI
DIAG
CURRENT-ADJUST
CUR-SENSE
LDO-SENSE
VIR-RTN
File: Strobe Generation.kicad_sch
PiTrac
Sheet: /
File: V3 Connector + IRLED.kicad_sch
Title: V3 Connector + IRLED
Size: A4
Date: 2026-02-12
Rev: V1
KiCad E.D.A. 9.0.8
Id: 1/7

Power Input
Screw Terminal 5V @ 5.1A+



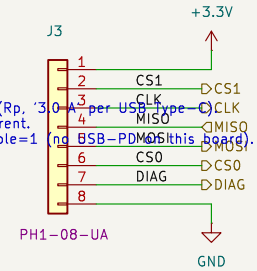
Raspberry Pi5 Power Only
USB-C 5V @ 5A





SPI & GPIO

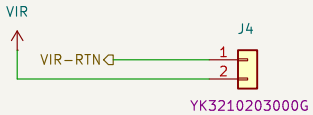
USB-C source advertisement (added).
J2/J5 supply 5 V to the Pi's so each CC pin gets a 10K pull-up to +5V (Rp, "3.0 A" per USB Type-C).
Without Rp a compliant sink may treat the port as unpowered/default-current.
A Pi 5 still limits USB peripherals to 600 mA unless usb_max_current_enable=1 (no USB-PD on this board).



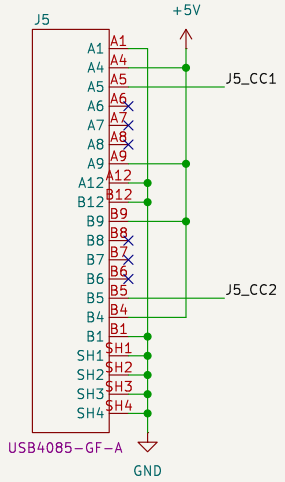
MOSI is STROBE



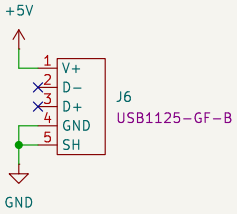
Strobe Output
Screw Terminal 36V @ 6 – 15A



Raspberry Pi5 Power Only
USB-C 5V @ 5A

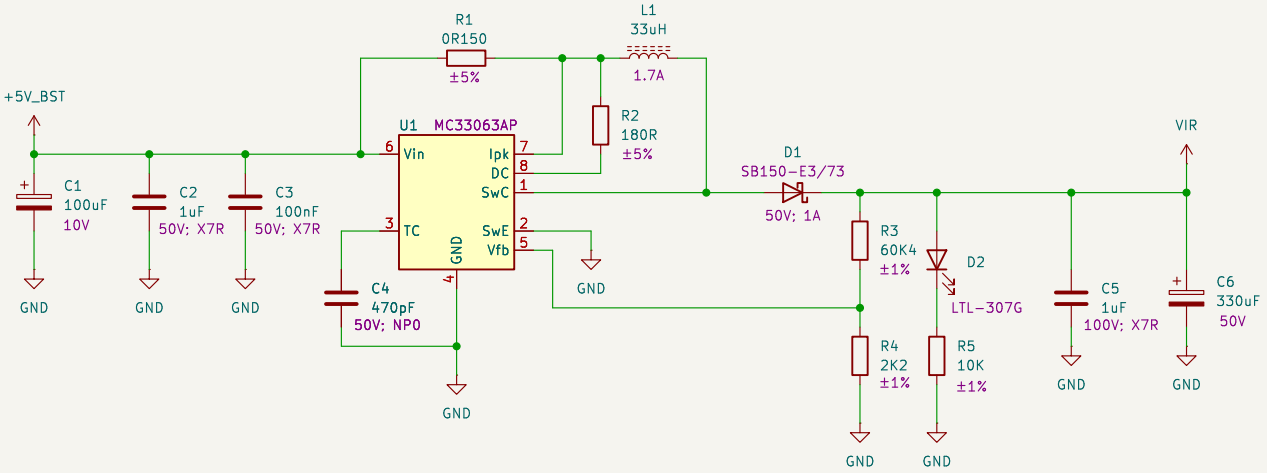


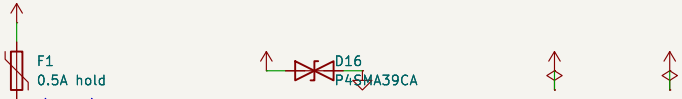
LED Strip (Optional)
USB-A 5V



PiTrac		
Sheet: /Connectors/		
File: Connectors.kicad_sch		
Title: V3 Connector + IRLED		
Size: A	Date: 2026-02-12	**Rev: V1**
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36V Boost Converter @ 0.1A

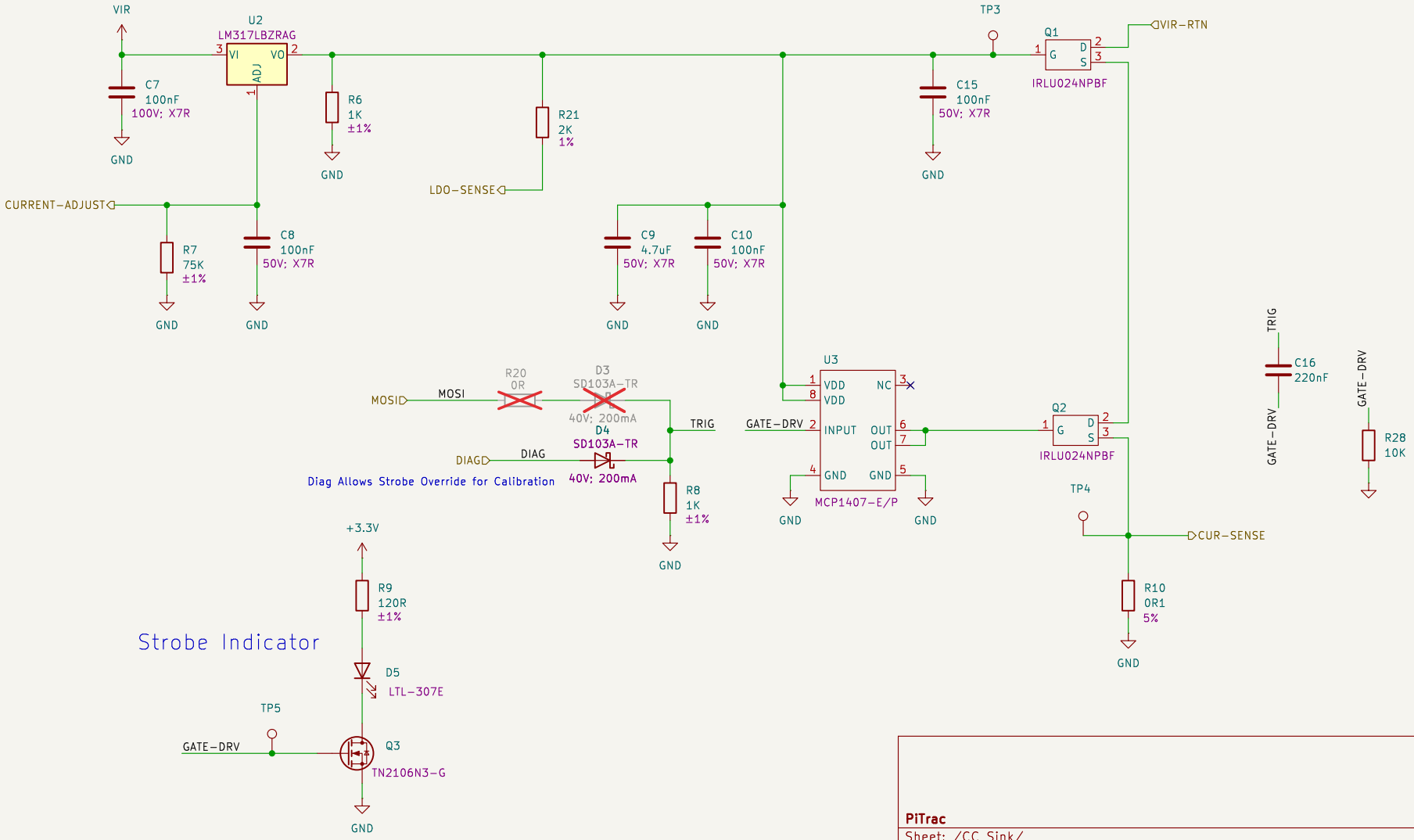




Protection (added).
F1: resettable fuse in the boost input only (not in the +5V path to the Pis). It limits a VIR short (e.g. C6, D16 or an LED/MOSFET failing short), which the MC33063 current limit cannot stop because current flows 5V > R1 > L1 > D1 directly.
D16: clamps VIR if the feedback divider (R3/R4) opens. VBR(min) 37.1 V is above worst-case normal VIR (~36.8 V). It is a clamp-then-fail-short scheme that relies on F1; the MC33063 (40 V) may not survive that fault, but the capacitors, LM317L, MOSFETs and LEDs are protected.

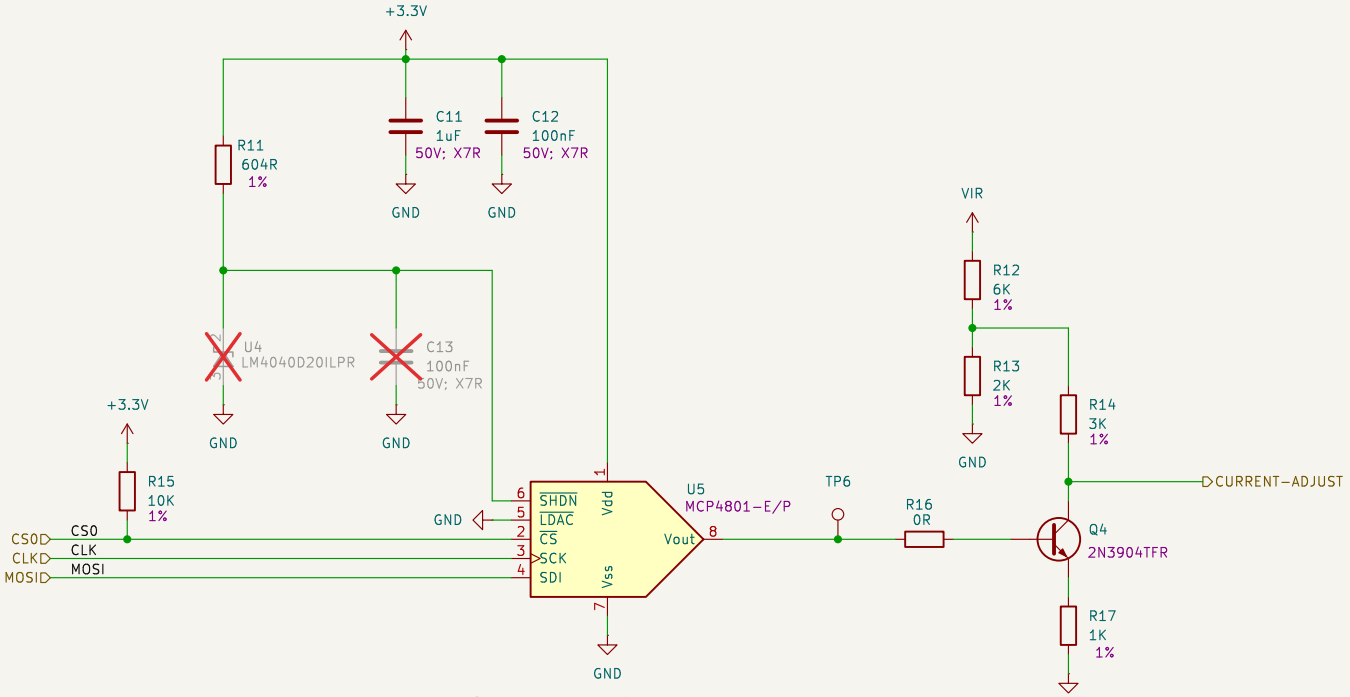
PiTrac		
Sheet: /Power/		
File: Power.kicad_sch		
Title: V3 Connector + IRLED		
Size: A4	Date: 2026-02-12	**Rev: V1**
KiCad E.D.A. 9.0.8		Id: 3/7

Adjustable Constant Current Sink



PiTrac		
Sheet: /CC Sink/		
File: Strobe Generation.kicad_sch		
Title: V3 Connector + IRLED		
Size: A4	Date: 2026-02-12	**Rev: V1**
KiCad E.D.A. 9.0.8		Id: 4/7

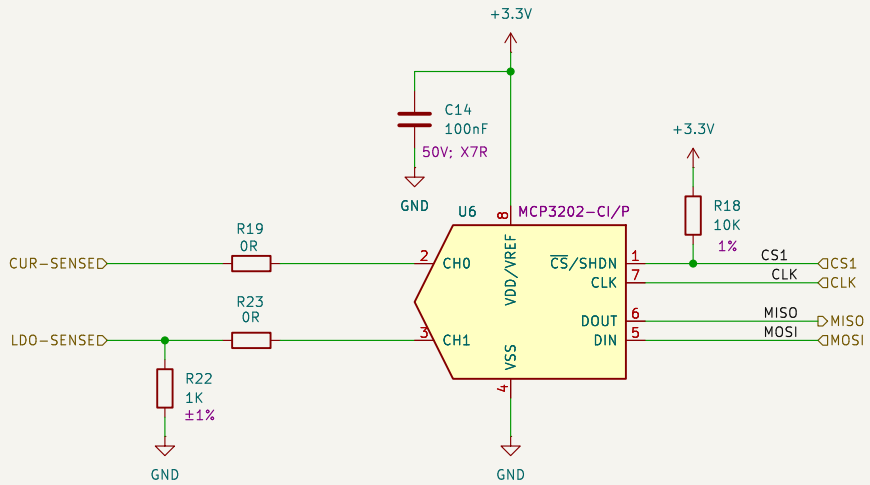
8-bit DAC w/ Ext or Int Ref



Current Adjust :
Control Range : 0.72 – 0.93V
Code Range : 5A – 74h
On Voltage : 4.5 – 8V
Voltage Resolution : ~135mV/count
Current Resolution : ~0.38A

PiTrac		
Sheet: /DAC/		
File: DAC.kicad_sch		
Title: V3 Connector + IRLED		
Size: A4	Date: 2026-02-12	**Rev: V1**
KiCad E.D.A. 9.0.8		Id: 5/7

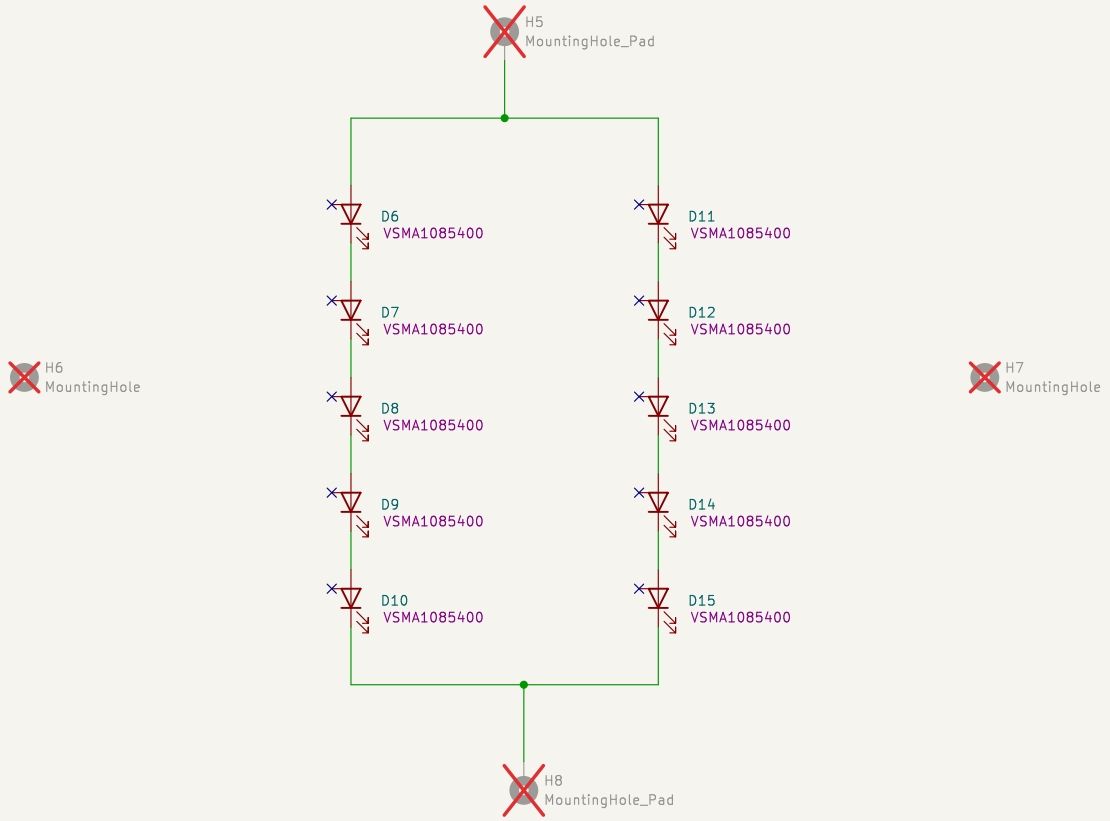
12-Bit ADC



Cur-Sense :
Voltage Resolution : ~800uV/count
Current = Voltage / 0.1
On Current : 5.5 - 15A
On Counts : 687 - 1862
Current Resolution : ~8mA/count

LDO-Sense :
Voltage Range : 1.5 - 2.6V

PiTrac		
Sheet: /ADC/ File: ADC.kicad_sch		
Title: V3 Connector + IRLED		
Size: A4	Date: 2026-02-12	**Rev: V1**
KiCad E.D.A. 9.0.8		Id: 6/7



PiTrac

Sheet: /LED/
File: IRLED.kicad_sch

Title: V3 Connector + IRLED

Size: A4	Date: 2026-02-12	**Rev: V1**
KiCad E.D.A. 9.0.8		Id: 7/7